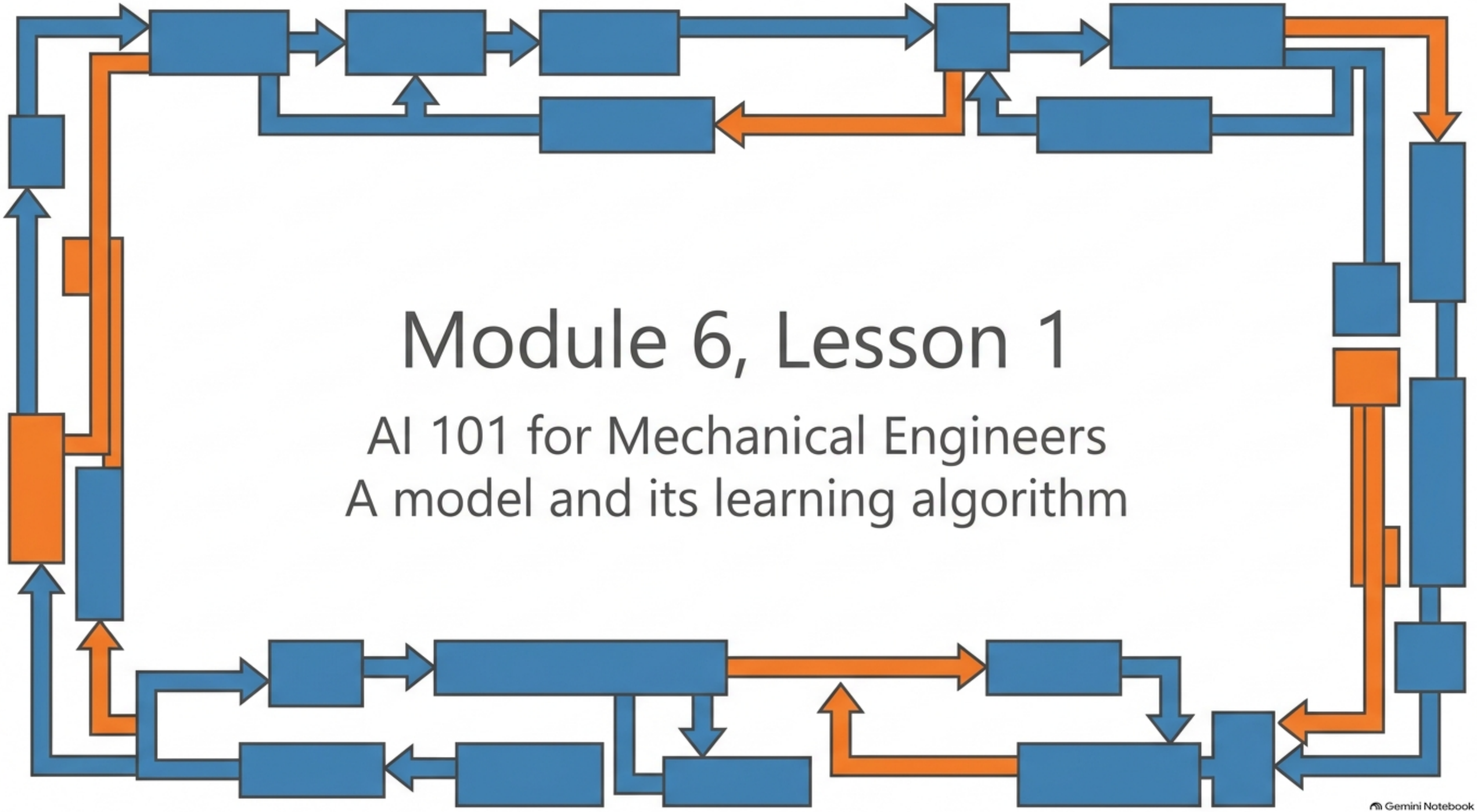


Module 6, Lesson 1

AI 101 for Mechanical Engineers

A model and its learning algorithm

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Course structural outline

1	**Architecture**
2	**Operation**
3	**Applied Linear System**
4	**Synthesis and Summary**

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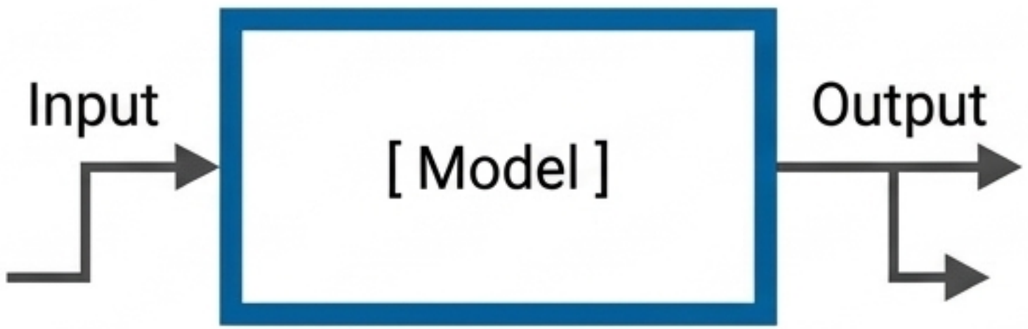
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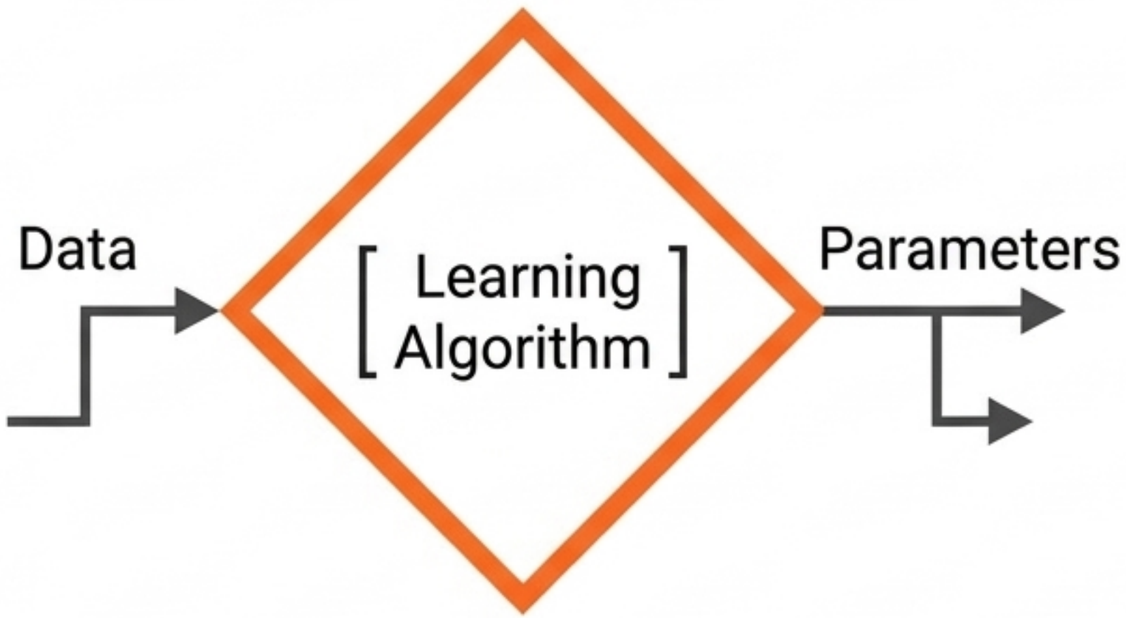
Section 1: Architecture

Distinguish the mathematical representation from the optimization procedure.

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System block definitions





Model architectures map inputs to mathematical outputs.

Learning algorithms extract system parameters from data.

Distinct algorithmic procedures fit identical structural models.

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2

Section 2: Operation

Define the distinct operational states of an artificial intelligence system.

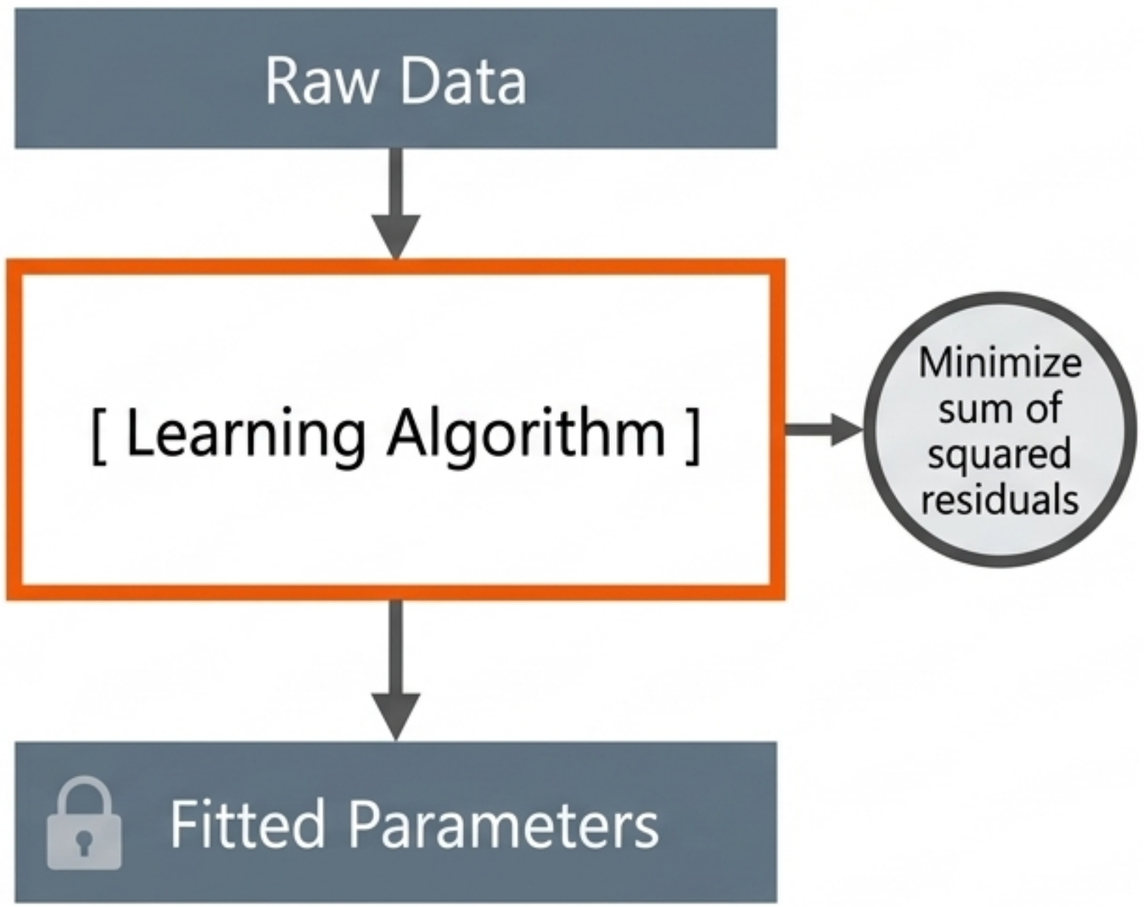
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Parameter estimation via training

Training extracts required parameter values from examples.

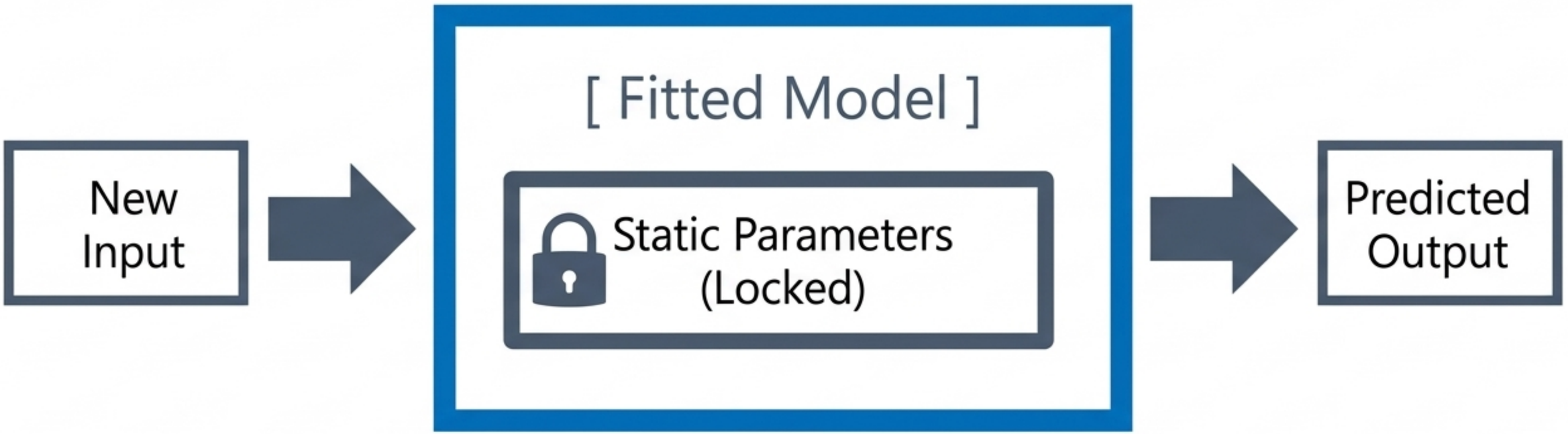
Fitting procedures execute specific optimization algorithms.

Least squares minimizes summed squared residual values.



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Output prediction via inference



Inference applies fitted models to novel inputs.

Routine prediction guarantees static parameter coefficients.

Generating outputs never updates mathematical system models.

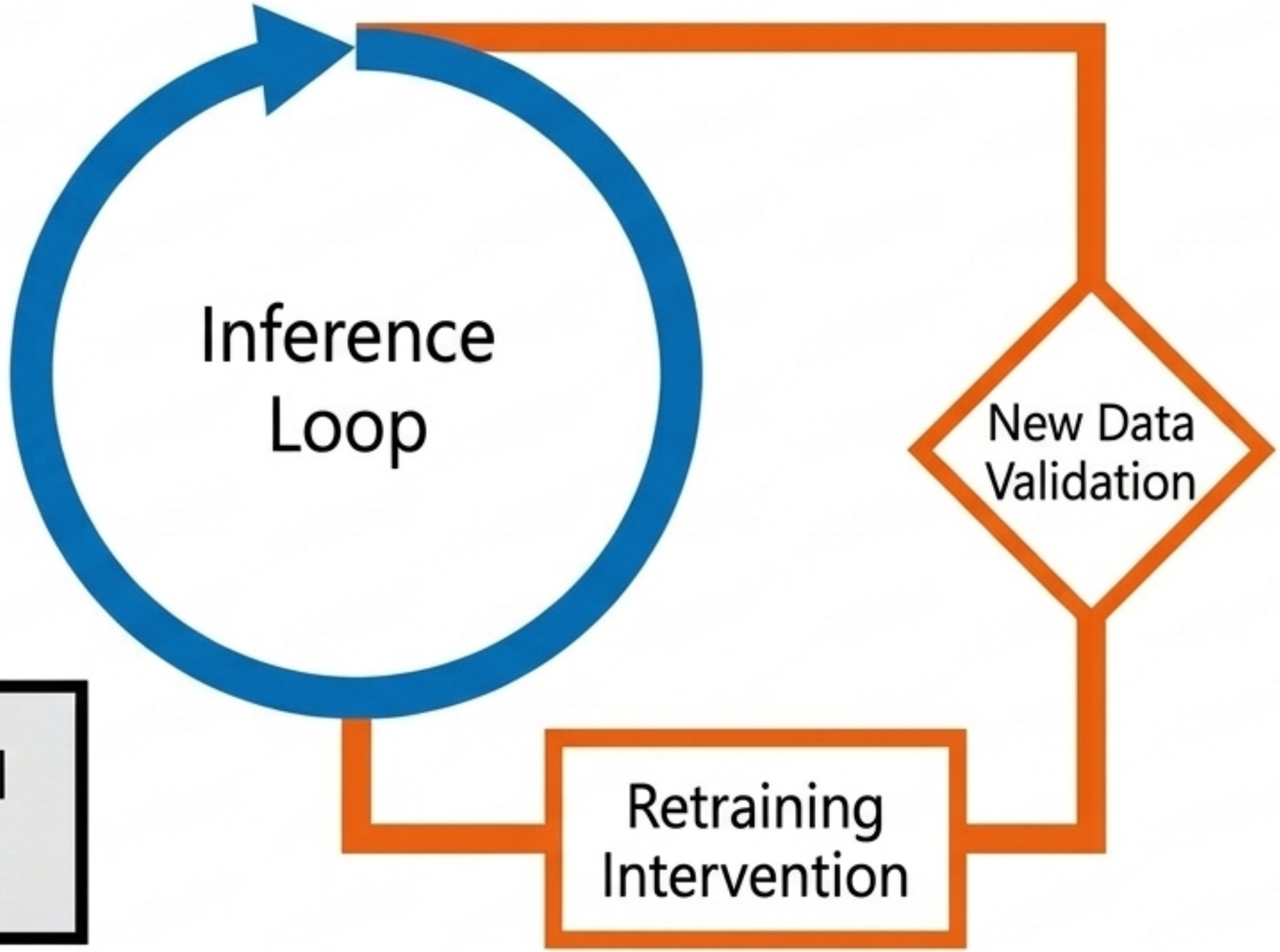
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System parameter updates via retraining

Retraining functions as an independent system operation.

Requires comprehensive acquisition of new operational data.

Mandates rigorous mathematical validation protocol execution.



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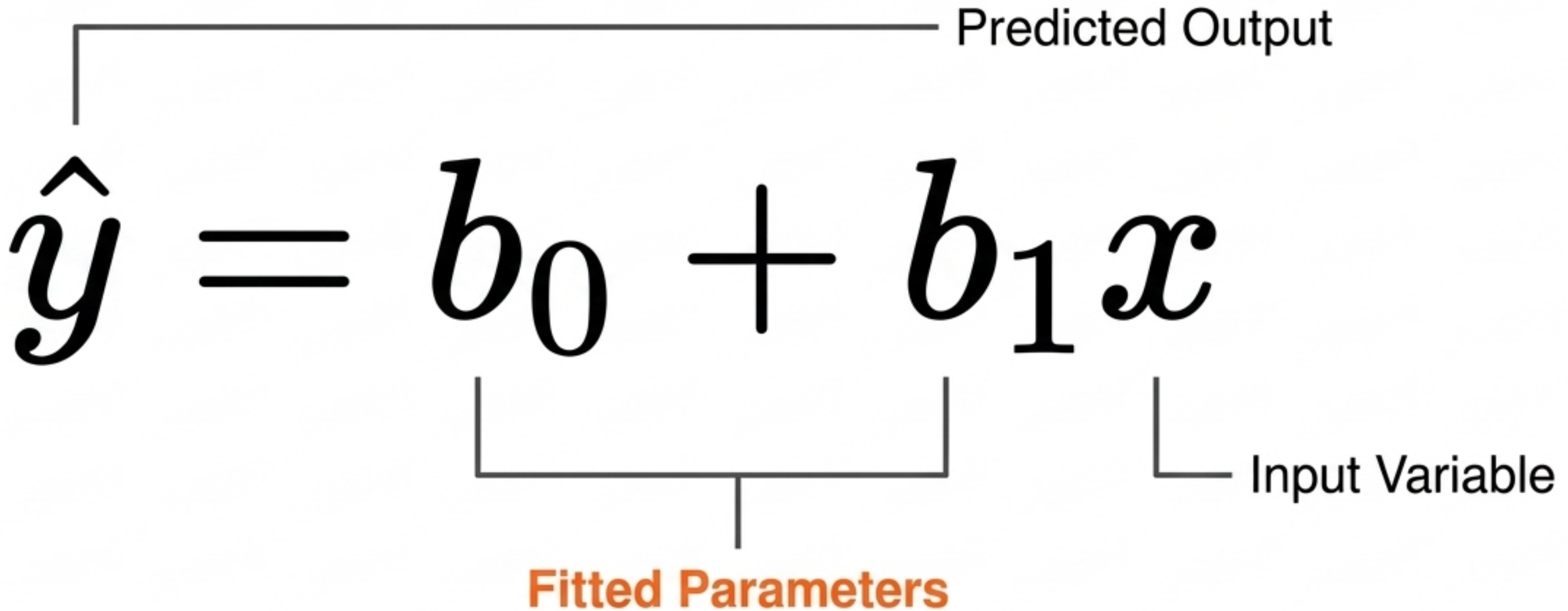
3

Section 3: Applied Linear System

Apply architectural concepts to a first-order linear representation.

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First-order linear equation parameters



Equation defines specific mathematical mapping relationships.	Coefficients function as static optimization targets.	Variables represent discrete system inputs and outputs.

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Inference execution for fixed parameters

Step 1: $\hat{y} = 2 + 3x$
Step 2: Input $x = 4$
Step 3: $\hat{y} = 2 + 3(4) = 14$ [Output units]

Fitted **intercept** parameter **strictly** equals **two**.

Fitted **slope** parameter strictly equals **three**.

Executing inference at input **four** yields **fourteen**.

Adjusting **input values** alters corresponding **output** calculations.

Internal system parameters remain **rigidly locked**.

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Section 4: Synthesis and Summary

Consolidate system vocabulary and distinct operational states.

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Synthesis of operational states

Operation	Parameter Status	Applied Linear Example
Training	Fitted coefficients	Estimate intercept and slope
Inference	New input and output	At $x = 4$, $2 + 3(4) = 14$
Retraining	A newly evaluated fit	Use new examples and validate

Matrix defines internal system variable transitions.
Delineates static versus dynamic mathematical phases.

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Key Points

1. Architecture distinguishes mathematical models from optimization algorithms.

2. Operation delineates static inference from dynamic parameter training.

3. Applied linear systems map discrete inputs to predictive outputs.

4. Synthesis matrix isolates parameter variability across operational modes.

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